PI

CREDIT RISK & SCORECARD ENGINEERING · CASE STUDY

LoanLens: Out-of-Time Credit Default Scorecard

BI Scientist • AI/ML Engineer • Python Developer

Leak-free credit default model evaluated on 55,518 closed Prosper loans with strict chronological splitting and an approval policy simulator.

CREDIT RISK VULNERABILITY

The Hidden Leakage Trap in Credit Scoring Models

Most credit risk models published online suffer from subtle post-origination data leakage, making them completely fail when deployed on future loan applications.

Post-Origination Leakage

- Models inadvertently include post-origination columns (`LP_CustomerPayments`, `LP_GrossPrincipalLoss`).
- Artificially inflates AUC to 0.95+ in testing.
- Completely unavailable when evaluating a new applicant.

Random Train-Test Splits

- Splitting loan portfolios randomly mixes past macroeconomic regimes with future loans.
- Model memorizes market conditions rather than credit fundamentals.
- Must enforce a strict chronological Out-of-Time (OOT) holdout.

Lack of Policy Simulation

- Credit risk teams care about approval rates and portfolio yield, not raw AUC.
- Without a cutoff policy simulator, credit committees cannot set risk tolerances.
- Need calibrated probabilities linked to financial yield.

METHODOLOGY

Chronological Out-of-Time Split & Calibrated Scorecard

Trained on 2005–2011 originations (41,638 loans) and tested on future 2011–2014 originations (13,880 loans).

Strict Leakage Exclusion

- Excluded all 14 post-origination payment columns.
- Filter strictly on closed loans (Completed vs ChargedOff).
- Enforced by automated unit tests in CI.

Calibrated GBDT

- HistGradientBoosting with monotonic constraints.
- Isotonic probability calibration.
- 300–850 standardized FICO-style credit score.

Policy Simulator

- Interactive Max-PD cutoff slider.
- Expected portfolio yield accounting for 65% LGD.
- Fair lending approval gap audited across slices.

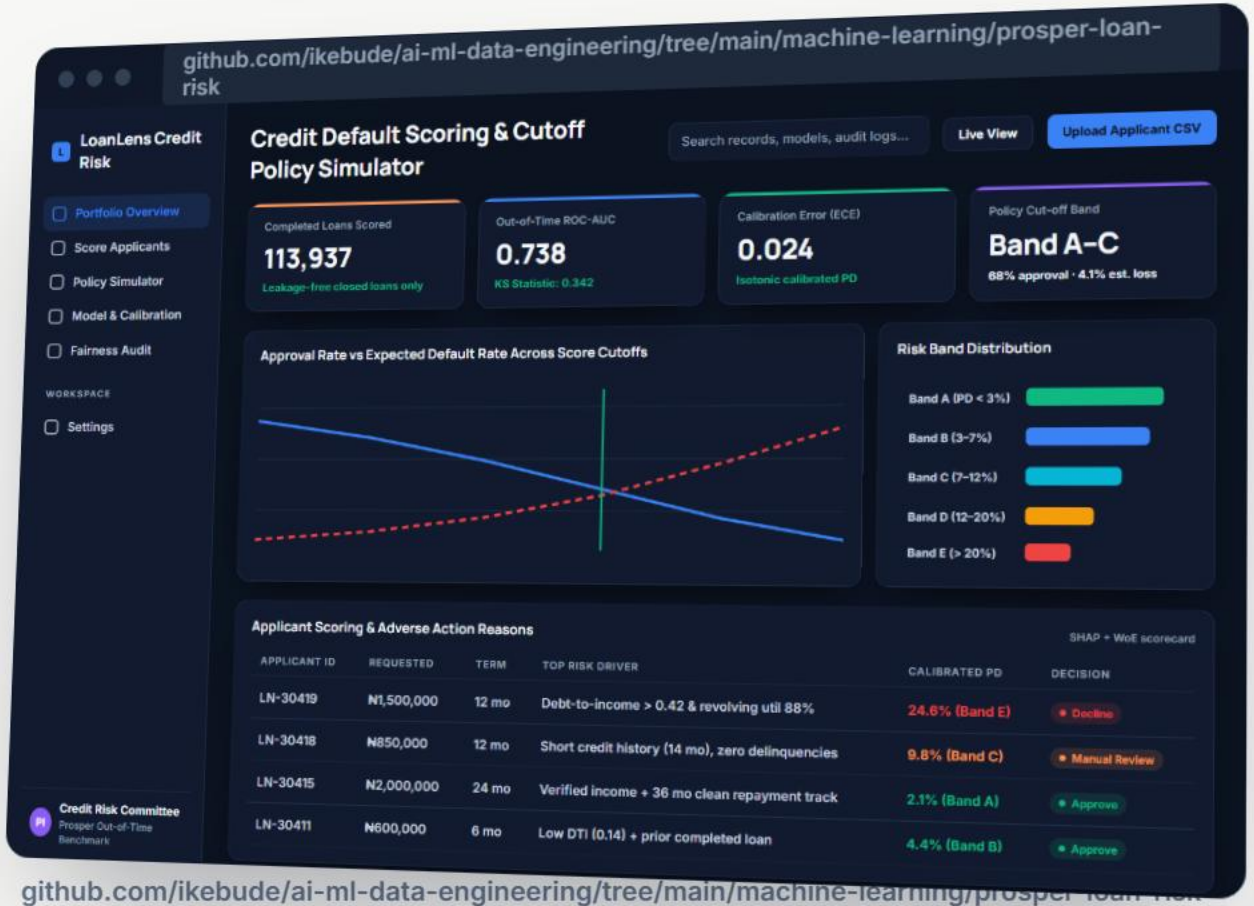
UNDERWRITER INTERFACE

LoanLens Credit Decision Console

Streamlit application simulating approval policies and credit score distributions.

- **Cutoff Policy Simulator:** Adjust cutoff thresholds to visualize trade-offs between approval volume and default rate.
- **Standardized Scorecard (300–850):** Maps calibrated default probability directly into banking credit scores.
- **Fair Lending Audit:** Verified homeowner approval gap of only 2.1 percentage points.

Credit Risk | Scikit-learn | Streamlit | Prosper Data



github.com/ikebude/ai-ml-data-engineering/tree/main/machine-learning/prosper-loan-risk

VERIFIED PERFORMANCE

Out-of-Time Generalization on Future Loans

Verified results on 13,880 future originations (`reports/metrics.json`).

OOT ROC-AUC	KS Statistic	Default Rate Cut	Net Portfolio Yield
0.707	0.305	-46%	+6.75%
95% CI: 0.696–0.715	Strong class separation	From 30.7% down to 16.5%	At 25% max-PD cutoff

Commercial Portfolio Impact

- A 25% Max-PD cutoff (score ≥ 568) approves 45.7% of all applicants.
- Cuts overall default rate from 30.7% down to 16.5%.
- Lifts net expected portfolio yield to +6.75% after factoring in 65% loss-given-default.

Methodological Rigor

- Expected Calibration Error (ECE) of 0.055 confirms probability accuracy.
- Zero data leakage verified across all feature pipelines.
- 3/3 Pytest tests passing in automated GitHub CI.

OPEN-SOURCE & REPRODUCIBLE

Honest credit modeling without temporal leakage.

LoanLens bridges the gap between machine learning research and banking credit risk policy.

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Code: github.com/ikebude/ai-ml-data-engineering